A HOMOLOGICAL RESOLUTION OF SIGNED ROUGH GOLDBACH CORRELATIONS OVER THE INTEGERS

Abstract. We give an integer-ring counterpart of a signed rough-factorization statistic that arises naturally in large-finite-field models of the Goldbach problem. For each additive representation $N = p + m$, the small-prime incidence set of m defines an augmented Koszul, equivalently bar, complex. After tensoring with a parity line, its Euler characteristic is

$$\sum_p \mu(N-p)\mathbf{1}_{P^-(N-p)>z}.$$

The associated incidence spectral sequence is computed explicitly: its fibres are augmented simplices, and it degenerates at E_2.

For $z = X^\theta$, a second filtration by the number of large prime factors has finitely many strata. Repeated use of the prime number theorem gives simplex integrals $I_j(\theta)$ for their archimedean main terms. We prove the exact identity

$$\sum_{j\geq 1}(-1)^j I_j(\theta) = -\rho(\theta^{-1}-1),$$

where ρ is the Dickman–de Bruijn function. This is the continuous counterpart of the symmetric-group coefficient in the function-field model. It yields an unconditional smoothed mean theorem. We also give a circle-method reduction, using small-modulus equidistribution and a large-sieve mean-square estimate, which yields the corresponding asymptotic for almost all even integers. Finally, we isolate the pointwise minor-arc estimate that would be needed to remove the exceptional set. At $\theta = 1/3$, its negativity would imply the binary Goldbach conjecture. The homological resolution therefore localizes, but does not remove, the parity obstruction.

Contents

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1. Introduction

1.1. **The signed rough correlation.** For an integer $m > 1$, write $P^{-}(m)$ for its least prime factor, with the convention $P^{-}(1) = +\infty$. Let N be an even integer and let $0 < \theta < 1$. The basic statistic considered in this note is

$$\Delta_\theta(N) := \sum_{\substack{p \le N/2 \\ p \text{ prime}}} \mu(N-p)\mathbf{1}_{P^{-}(N-p) > N^\theta}. \tag{1}$$

For squarefree m, the value $\mu(m) = (-1)^{\omega(m)}$ records the parity of the number of prime factors. Thus $\Delta_\theta(N)$ is the number of rough targets with an even number of factors minus the number with an odd number of factors.

There are three distinct questions.

(1) Can the roughness indicator and the parity sign be encoded by a genuinely homological object?
(2) What is the signed archimedean main term?
(3) How much of that main term survives after imposing the additive prime condition $N = p+m$?

The first two questions admit exact answers. The third has an unconditional mean and density-one answer, whereas a pointwise answer reaches the binary Goldbach problem.

1.2. **Main conclusions.** The finite-set homological resolution is stated in Theorem 2.2. Its incidence spectral sequence is computed in Proposition 2.4. The main archimedean coefficient is the following.

Theorem 1.1 (Signed Dickman coefficient)**.** *For $0 < \theta < 1$, define*

$$I_1(\theta) = 1$$

and, for $j \ge 2$,

$$I_j(\theta) := \frac{1}{j!} \int_{\substack{u_i > \theta \\ u_1 + \cdots + u_j = 1}} \frac{\mathrm{d}u_1 \cdots \mathrm{d}u_{j-1}}{u_1 \cdots u_j}.$$

The integral is zero when $j\theta \ge 1$. Then

$$\sum_{j \ge 1} (-1)^j I_j(\theta) = -\rho\left(\frac{1-\theta}{\theta}\right), \tag{2}$$

where $\rho(u) = 1$ *for* $0 \le u \le 1$ *and*

$$u\rho'(u) + \rho(u-1) = 0 \qquad (u > 1).$$

In particular, the signed coefficient is strictly negative.

Theorem 1.1 is proved in Section 4. It also explains the exact large-degree limit of the function-field permutation coefficient; see Section 5.

For a clean analytic statement, let $w_1, w_2 \in C_c^\infty((0,\infty))$, put

$$b_{\theta,X}(m) := \mu(m)\mathbf{1}_{P^-(m) > X^\theta}, \tag{3}$$

and define

$$\mathcal{G}_{\theta,\mathbf{w}}(N;X) := \sum_{\substack{p+m=N \\ p \text{ prime}}} w_1(p/X)w_2(m/X)b_{\theta,X}(m). \tag{4}$$

Set

$$\mathcal{J}_{\mathbf{w}}(\nu) := \int_{\mathbb{R}} w_1(t)w_2(\nu - t)\,\mathrm{d}t. \tag{5}$$

The standard Goldbach singular series is

$$\mathfrak{S}(N) := \sum_{q=1}^{\infty} \frac{\mu(q)^2}{\varphi(q)^2} c_q(-N), \tag{6}$$

where $c_q(n)$ is the Ramanujan sum. It is positive for even N.

Theorem 1.2 (Density-one signed rough Goldbach asymptotic)**.** *Fix* $0 < \theta < 1$ *and* $w_1, w_2 \in C_c^\infty((0,\infty))$*. For all but* $o(X)$ *even integers* $N \in [X, 2X]$*,*

$$\mathcal{G}_{\theta,\mathbf{w}}(N;X) = -\rho(\theta^{-1} - 1)\mathfrak{S}(N)\frac{X}{(\log X)^2}\mathcal{J}_{\mathbf{w}}(N/X) + o_{\theta,\mathbf{w}}\left(\frac{X}{(\log X)^2}\right). \tag{7}$$

We give the standard circle-method proof in Section 7. The only analytic inputs are Siegel–Walfisz estimates for finitely many prime-factor strata and a Vaughan-type minor-arc estimate for the prime exponential sum. We include the proof to show exactly where averaging over N is used.

1.3. **The pointwise barrier.** At $\theta = 1/3$, a number $m < N$ with $P^-(m) > N^{1/3}$ has at most two prime factors. Consequently

$$\Delta_{1/3}(N) = S(N) - R(N), \tag{8}$$

where $R(N)$ counts representations $N = p + q$ with $p \le N/2$, and $S(N)$ counts representations

$$N = p + rs, \qquad r, s > N^{1/3},$$

with $r \ne s$. Hence

$$\Delta_{1/3}(N) < 0 \quad \Longrightarrow \quad R(N) > 0.$$

A pointwise negative asymptotic would therefore prove binary Goldbach. Section 9 identifies the missing estimate as a pointwise minor-arc, or twisted dispersion, bound. The role of the homological resolution is diagnostic: it isolates the obstruction in the first prime-factor stratum.

1.4. **Status and scope.** The complexes, the spectral-sequence computation, the simplex-integral identity, and the smoothed mean theorem are proved directly here. The density-one theorem uses standard analytic inputs, written in a form adapted to the factor strata. No pointwise Goldbach claim is made. We also make no claim that every individual analytic ingredient is new. The potentially useful contribution is the common framework:

$$\text{bar/Koszul resolution} \longrightarrow \text{factor-number filtration}$$
$$\longrightarrow \text{Dickman coefficient} \longrightarrow \text{circle-method correlation.}$$

2. The integer homological sieve complex

2.1. **Parity lines.** For a squarefree integer m, let

$$L_m := \mathbb{Q}[\omega(m)] \tag{9}$$

be a one-dimensional complex placed in cohomological degree $\omega(m)$. For nonsquarefree m, put $L_m = 0$. In the Grothendieck group of finite complexes,

$$\chi(L_m) = \mu(m). \tag{10}$$

One may replace $\mathbb{Q}[\omega(m)]$ by the determinant line of the vector space spanned by the prime factors of m; only its cohomological shift is used below.

2.2. **The small-prime incidence Koszul complex.** For $z \geq 2$, set

$$S_z(m) := \{\ell \leq z : \ell \mid m,\ \ell \text{ prime}\}$$

and let $V_z(m)$ be the vector space with basis $S_z(m)$. If $S_z(m) \neq \varnothing$, let

$$\varepsilon_m : V_z(m) \longrightarrow \mathbb{Q}$$

send every basis vector to 1. Define

$$K_z(m) := \left(\bigwedge\nolimits^{\bullet} V_z(m), \iota_{\varepsilon_m}\right), \tag{11}$$

where ι_{ε_m} is contraction by ε_m. If $S_z(m) = \varnothing$, define $K_z(m) = \mathbb{Q}$ in degree zero.

Lemma 2.1. *One has*

$$H^{\bullet}(K_z(m)) \cong \begin{cases} \mathbb{Q}, & P^-(m) > z, \\ 0, & P^-(m) \leq z. \end{cases}$$

Proof. If $V_z(m) = 0$, the assertion follows from the definition. Otherwise choose $v \in V_z(m)$ with $\varepsilon_m(v) = 1$. Exterior multiplication by v is a contracting homotopy:

$$\iota_{\varepsilon_m}(v \wedge \eta) + v \wedge \iota_{\varepsilon_m}(\eta) = \eta.$$

□

Let

$$\mathcal{P}_N := \{p \leq N/2 : p \text{ prime}\}$$

and define the total finite complex

$$\mathcal{C}_{N,z} := \bigoplus_{p \in \mathcal{P}_N} L_{N-p} \otimes K_z(N-p). \tag{12}$$

Theorem 2.2 (Homological resolution)**.** *The cohomology and Euler characteristic of $\mathcal{C}_{N,z}$ are*

$$H^\bullet(\mathcal{C}_{N,z}) \cong \bigoplus_{\substack{p\in\mathcal{P}_N \\ P^-(N-p)>z}} L_{N-p}, \tag{13}$$

$$\chi(\mathcal{C}_{N,z}) = \sum_{p\in\mathcal{P}_N} \mu(N-p)\mathbf{1}_{P^-(N-p)>z}. \tag{14}$$

Proof. Apply Lemma 2.1 to each direct summand and then use (10). □

Remark 2.3. This construction is an integer, finite-set shadow of the bar-complex inclusion–exclusion used in geometric homological sieves; compare [9]. Unlike the geometric situation, there is no Frobenius action or weight theorem attached to the finite set $\mathcal{P}_N$. The complex gives an exact resolution, not an automatic cancellation estimate.

2.3. **The incidence spectral sequence.** Let $\mathcal{Q}(z)$ denote the set of primes at most z. The bar model of $K_z(N-p)$ has r-simplices indexed by subsets

$$J = \{\ell_0 < \cdots < \ell_r\} \subseteq S_z(N-p).$$

After summing over p, the simplicial filtration gives an E_1-page whose trace-weighted terms are

$$E_1^{-r,*} = \bigoplus_{\substack{J\subseteq\mathcal{Q}(z) \\ |J|=r+1}} \bigoplus_{\substack{p\in\mathcal{P}_N \\ d_J\mid N-p}} L_{N-p}, \qquad d_J := \prod_{\ell\in J}\ell. \tag{15}$$

The differential is the alternating sum of maps deleting one prime from J.

Proposition 2.4 (Spectral-sequence computation)**.** *For a fixed $p\in\mathcal{P}_N$, the fibre of* (15) *is the augmented simplicial chain complex on $S_z(N-p)$. Hence its E_2-page is*

$$E_2(p) = \begin{cases} L_{N-p}, & P^-(N-p) > z, \\ 0, & P^-(N-p) \le z. \end{cases}$$

The global spectral sequence degenerates at E_2 and abuts to (13)*.*

Proof. If $S_z(N-p)\neq\varnothing$, the augmented simplex on that vertex set is contractible. If the set is empty, only the augmentation term remains. There are no additional geometric cohomological degrees, so no differential d_r, $r\ge 2$, can survive. □

2.4. **Truncation and residual homology.** Analytically, one can estimate the terms in (15) only for a restricted collection of moduli d_J, typically $d_J \le D$. Because deleting a prime decreases d_J, the subsets with $d_J\le D$ form a downward-closed subcomplex. The discarded cells contribute the relative complex

$$K_z(N-p)/K_{z,\le D}(N-p).$$

The full simplex is acyclic when $S_z(N-p)\neq\varnothing$, but its truncation need not be. Thus the sieve remainder is literally a relative homology group.

This observation does not prove a parity-breaking estimate. It identifies what such an estimate would have to do: control the trace of the residual homology beyond the available distribution level. At the classical square-root level $D\approx X^{1/2}$, this is a homological formulation of the usual parity barrier.

3. The factor-number filtration

Fix $0 < \theta < 1$ and consider $m \asymp X$. Up to a boundary of lower order, every squarefree m counted by $b_{\theta,X}$ can be written uniquely as

$$m = p_1 \cdots p_j, \qquad X^\theta < p_1 < \cdots < p_j,$$

with $j\theta < 1$. Hence

$$b_{\theta,X}(m) = \sum_{\substack{j\geq 1\\ j\theta<1}} (-1)^j b_{j,\theta,X}(m), \tag{16}$$

where $b_{j,\theta,X}$ is the indicator of the j-factor stratum. Repeated prime factors do not occur because of the Möbius weight.

The filtration by $\omega(m)$ is finite for every fixed θ. Its associated graded terms are the large-prime factorization strata. In Dirichlet-polynomial language,

$$\sum_m \frac{b_{j,\theta,X}(m)\chi(m)}{m^s} = \frac{1}{j!}\left(\sum_{p>X^\theta} \frac{\chi(p)}{p^s}\right)^j \text{diagonal and support corrections.} \tag{17}$$

This finite resolution is analytically preferable to treating the total Euler product $1/L(s,\chi)$ directly.

3.1. Archimedean densities of the strata.

For $j \geq 2$, write $u_j = 1 - u_1 - \cdots - u_{j-1}$ and define

$$I_j(\theta) := \frac{1}{j!}\int_{\substack{u_1,\ldots,u_j>\theta\\ u_1+\cdots+u_j=1}} \frac{\mathrm{d}u_1\cdots \mathrm{d}u_{j-1}}{u_1\cdots u_j}, \qquad I_1(\theta) := 1. \tag{18}$$

Lemma 3.1 (Stratum density)**.** *Let $w \in C_c^\infty((0,\infty))$. For fixed j and θ,*

$$\sum_m b_{j,\theta,X}(m)w(m/X) = I_j(\theta)\frac{X}{\log X}\int_0^\infty w(t)\,\mathrm{d}t O_{j,\theta,w}\left(\frac{X}{(\log X)^2}\right). \tag{19}$$

The same leading coefficient holds in a reduced residue class modulo $q \leq (\log X)^B$, with the main term divided by $\varphi(q)$.

Proof. Write $p_i = X^{u_i}$ and apply the prime number theorem successively to the j prime variables. The local measure contributed by a prime p_i is

$$\frac{\mathrm{d}p_i}{\log p_i} = \frac{X^{u_i}\,\mathrm{d}u_i}{u_i},$$

while the product constraint contributes the overall factor $X/\log X$. Division by $j!$ removes ordering. This gives (18). Repeated primes and boundary faces have lower order.

For the progression statement, every prime divisor of $q \leq (\log X)^B$ is less than X^θ, so every counted m is automatically coprime to q. Fixing $j-1$ prime variables leaves the final prime in a reduced residue class modulo q. Siegel–Walfisz and partial summation give uniform equidistribution. Repeated Taylor expansion of the logarithmic densities gives a full expansion in powers of $1/\log X$, if needed on major arcs. ☐

Corollary 3.2. *For $w \in C_c^\infty((0,\infty))$,*

$$\sum_m b_{\theta,X}(m)w(m/X) = C(\theta)\frac{X}{\log X}\int_0^\infty w(t)\,\mathrm{d}t + O_{\theta,w}\left(\frac{X}{(\log X)^2}\right), \tag{20}$$

where

$$C(\theta) := \sum_{j\geq 1}(-1)^j I_j(\theta).$$

4. The signed Dickman identity

The finite alternating sum in Corollary 3.2 has a closed form.

Lemma 4.1 (Boundary recurrence)**.** *For $j \geq 2$,*

$$I_j'(\theta) = -\frac{1}{\theta(1-\theta)} I_{j-1}\left(\frac{\theta}{1-\theta}\right). \tag{21}$$

Proof. Increasing θ moves each of the j boundary faces $u_i = \theta$. By symmetry, the factor j cancels the $j!$ in (18). On the face $u_1 = \theta$, the remaining variables sum to $1-\theta$. Put

$$u_i = (1-\theta)v_i, \qquad i = 2, \dots, j.$$

The lower bound becomes $v_i > \theta/(1-\theta)$, and the measure divided by the product of the variables contributes the factor $1/(1-\theta)$. The fixed boundary variable contributes $1/\theta$. This gives (21). □

Theorem 4.2 (Signed Dickman identity)**.** *For $0 < \theta < 1$,*

$$C(\theta) = -\rho\left(\frac{1-\theta}{\theta}\right). \tag{22}$$

Proof. Lemma 4.1 gives

$$\begin{aligned} C'(\theta) &= \sum_{j \geq 2} (-1)^j I_j'(\theta) \\ &= \frac{1}{\theta(1-\theta)} C\left(\frac{\theta}{1-\theta}\right). \end{aligned}$$

Define

$$D(u) := -C\left(\frac{1}{u+1}\right), \qquad u \geq 0.$$

If $0 \leq u \leq 1$, then $1/(u+1) \geq 1/2$; only I_1 is nonzero, so $D(u) = 1$. For $u > 1$, use

$$\theta = \frac{1}{u+1}, \qquad \frac{\theta}{1-\theta} = \frac{1}{u} = \frac{1}{(u-1)+1}.$$

The chain rule gives

$$D'(u) = -\frac{D(u-1)}{u}.$$

Thus D satisfies the defining delay differential equation and initial condition of the Dickman–de Bruijn function. Uniqueness gives $D(u) = \rho(u)$, proving (22). □

4.1. **Explicit ranges.** For $1/2 \leq \theta < 1$, only the prime stratum occurs:

$$C(\theta) = -1.$$

For $1/3 \leq \theta < 1/2$,

$$I_2(\theta) = \frac{1}{2}\int_\theta^{1-\theta} \frac{\mathrm{d}u}{u(1-u)} = \log\frac{1-\theta}{\theta},$$

and hence

$$C(\theta) = -1 + \log\frac{1-\theta}{\theta}. \tag{23}$$

For $1/4 \leq \theta < 1/3$, a third stratum occurs, with

$$I_3(\theta) = \frac{1}{3}\int_\theta^{1-2\theta} \frac{1}{u(1-u)} \log\frac{1-u-\theta}{\theta}\, \mathrm{d}u. \tag{24}$$

The first reciprocal thresholds are

θ	$(1-\theta)/\theta$	$C(\theta)$
$1/3$	2	$-(1-\log 2) \approx -0.30685282$
$1/4$	3	$-\rho(3) \approx -0.04860839$
$1/5$	4	$-\rho(4) \approx -0.00491093$
$1/6$	5	$-\rho(5) \approx -0.00035472.$

The rapid decay shows why smaller θ requires correspondingly sharper analytic error terms.

5. The function-field dictionary

Factor degrees of a random squarefree polynomial of degree n over a large finite field are modeled by cycle lengths of a random permutation in S_n. Suppose $y/n \to \theta$. Let $p_y(m)$ be the probability that a random permutation in S_m has all cycle lengths at most y. Then

$$m = n - y - 1 \sim (1-\theta)n, \qquad \frac{y}{m} \longrightarrow \frac{\theta}{1-\theta}.$$

The longest-cycle limit law for random permutations gives

$$p_y(n-y-1) \longrightarrow \rho\left(\frac{1-\theta}{\theta}\right); \tag{25}$$

see, for example, [4].

Thus a function-field signed coefficient of the form

$$-\frac{1}{n} p_y(n-y-1)$$

has the exact archimedean counterpart

$$-\frac{1}{\log X}\rho\left(\frac{1-\theta}{\theta}\right).$$

The dictionary is

$\mathbb{F}_q[T]$	$\mathbb{Z}$
$\deg F = n$	$\log X$
$\deg P/n$	$\log p/\log X$
$1/n$	$1/\log X$
cycle decomposition	prime factorization
$p_y(n-y-1)$	$\rho((1-\theta)/\theta).$

This is the precise sense in which Theorem 4.2 is the integer-ring limit of the symmetric-group calculation. Function-field equidistribution can additionally use monodromy and finite-field trace estimates; see, for example, [1]. There is no direct pointwise analogue of that geometric error estimate over $\mathbb{Z}$.

6. An unconditional smoothed mean

The first analytic consequence does not require the circle method.

Theorem 6.1 (Smoothed global mean)**.** *Fix $0 < \theta < 1$ and $W \in C_c^\infty((0,\infty)^2)$. Then*

$$\sum_{\substack{p \text{ prime} \\ m \geq 1}} W(p/X, m/X)\mu(m)\mathbf{1}_{P^-(m) > X^\theta}$$

$$= -\rho(\theta^{-1}-1)\frac{X^2}{(\log X)^2}\int_0^\infty\int_0^\infty W(s,t)\,\mathrm{d}s\,\mathrm{d}t + o_{\theta,W}\left(\frac{X^2}{(\log X)^2}\right). \tag{26}$$

Proof. Apply the prime number theorem with a smooth weight to the p-variable, and Corollary 3.2 to the m-variable. For a non-separable W, approximate W in the smooth topology by finite sums of tensor products. The compact support makes the approximation uniform. Theorem 4.2 supplies the coefficient. □

Remark 6.2. Theorem 6.1 is a genuine integer theorem, but its analytic content is standard prime and almost-prime counting. Its value is that the homological factor filtration produces the signed Dickman constant without an appeal to a heuristic independence model.

7. Circle method and the density-one theorem

7.1. **Fourier setup.** Let

$$S_\Lambda(\alpha) := \sum_n \Lambda(n) w_1(n/X) \mathrm{e}(n\alpha), \tag{27}$$

$$S_\theta(\alpha) := \sum_m b_{\theta,X}(m) w_2(m/X) \mathrm{e}(m\alpha), \tag{28}$$

where $\mathrm{e}(x) := \exp(2\pi i x)$. The logarithmically weighted correlation is

$$\mathcal{R}_{\theta,\mathbf{w}}(N;X) := \sum_{n+m=N} \Lambda(n) w_1(n/X) w_2(m/X) b_{\theta,X}(m) = \int_0^1 S_\Lambda(\alpha) S_\theta(\alpha) \mathrm{e}(-N\alpha)\,\mathrm{d}\alpha. \tag{29}$$

Choose $P = (\log X)^B$. The major arcs are

$$\mathfrak{M} := \bigcup_{\substack{q\le P\\ (a,q)=1}} \left\{ \alpha : \left|\alpha - \frac{a}{q}\right| \le \frac{P}{qX} \right\},$$

and $\mathfrak{m} = [0,1] \setminus \mathfrak{M}$.

7.2. **Small-modulus local model.**

Lemma 7.1 (Major-arc local model)**.** *Fix $B, K > 0$. Uniformly for $q \le (\log X)^B$, $(a,q) = 1$, and $|\beta| \le (\log X)^B/X$, the sum $S_\theta(a/q + \beta)$ has an expansion*

$$S_\theta\left(\frac{a}{q} + \beta\right) = \frac{\mu(q)}{\varphi(q)} \frac{X}{\log X} \left[-\rho(\theta^{-1} - 1)\widehat{w}_{2,\beta} + \sum_{\nu=1}^{K-1} \frac{\mathcal{B}_{\nu,\theta,w_2}(\beta X)}{(\log X)^\nu} \right] + O\left(\frac{X}{(\log X)^{K+1}} \right), \tag{30}$$

where

$$\widehat{w}_{2,\beta} := \int_0^\infty w_2(t) \mathrm{e}(\beta X t)\,\mathrm{d}t,$$

and the secondary coefficient functions are smooth and rapidly decreasing.

Proof. Insert the finite decomposition (16). For each fixed stratum, fix all but the final prime variable and use Siegel–Walfisz in the resulting reduced residue class modulo q. The product of independent uniform elements of $(\mathbb{Z}/q\mathbb{Z})^\times$ is uniform, and

$$\frac{1}{\varphi(q)} \sum_{\substack{r \bmod q\\ (r,q)=1}} \mathrm{e}(ar/q) = \frac{\mu(q)}{\varphi(q)}.$$

Taylor expansion of the logarithmic densities gives the stated expansion. Its leading signed coefficient is Theorem 4.2. □

The corresponding standard prime approximation is

$$S_\Lambda\left(\frac{a}{q}+\beta\right)=\frac{\mu(q)}{\varphi(q)}X\widehat{w}_{1,\beta}+O_{A,w_1}\left(\frac{X}{(\log X)^A}\right), \tag{31}$$

for every fixed A, where

$$\widehat{w}_{1,\beta}=\int_0^\infty w_1(t)\mathrm{e}(\beta Xt)\,\mathrm{d}t.$$

Proposition 7.2 (Major arcs)**.** *Uniformly for even $N\in[X,2X]$,*

$$\begin{aligned}&\int_{\mathfrak{M}} S_\Lambda(\alpha)S_\theta(\alpha)\mathrm{e}(-N\alpha)\,\mathrm{d}\alpha\\&\qquad=-\rho(\theta^{-1}-1)\mathfrak{S}(N)\frac{X}{\log X}\mathcal{J}_{\mathbf{w}}(N/X)+O_{\theta,\mathbf{w}}\left(\frac{X}{(\log X)^2}\right).\end{aligned} \tag{32}$$

Proof. Insert (30) and (31), choosing K large relative to B. Summation over reduced a mod q produces $c_q(-N)$, and the q-sum gives the truncated version of (6). Absolute convergence supplies the tail. Fourier inversion in β gives $\mathcal{J}_{\mathbf{w}}(N/X)$. The first omitted logarithmic term is $O(X/(\log X)^2)$. □

7.3. **Minor arcs in mean square.**

Lemma 7.3 (Mean-square minor arcs)**.** *For every $A>0$, one can choose $B=B(A,\theta,\mathbf{w})$ such that*

$$\sum_{N\in\mathbb{Z}}\left|\int_{\mathfrak{m}} S_\Lambda(\alpha)S_\theta(\alpha)\mathrm{e}(-N\alpha)\,\mathrm{d}\alpha\right|^2\ll_{A,\theta,\mathbf{w}}\frac{X^3}{(\log X)^A}. \tag{33}$$

Proof. A Vaughan-type estimate for the prime exponential sum gives

$$\sup_{\alpha\in\mathfrak{m}}|S_\Lambda(\alpha)|\ll_{A,\mathbf{w}}\frac{X}{(\log X)^{A+2}}$$

after taking B sufficiently large; see, for example, [5, Chapters 13 and 19] or [6]. Parseval and Cauchy–Schwarz give

$$\begin{aligned}&\sum_N\left|\int_{\mathfrak{m}} S_\Lambda(\alpha)S_\theta(\alpha)\mathrm{e}(-N\alpha)\,\mathrm{d}\alpha\right|^2\\&\qquad\le\sup_{\alpha\in\mathfrak{m}}|S_\Lambda(\alpha)|^2\int_0^1|S_\theta(\alpha)|^2\,\mathrm{d}\alpha.\end{aligned}$$

Finally,

$$\int_0^1|S_\theta(\alpha)|^2\,\mathrm{d}\alpha=\sum_m|b_{\theta,X}(m)|^2|w_2(m/X)|^2\ll_{\theta,w_2}\frac{X}{\log X},$$

by the usual upper bound for X^θ-rough integers. Enlarging A proves (33). □

Proof of Theorem 1.2. Proposition 7.2 gives the main term. By Lemma 7.3 and Chebyshev's inequality, the minor-arc term is $o(X/\log X)$ for all but $o(X)$ integers $N\in[X,2X]$. Hence

$$\mathcal{R}_{\theta,\mathbf{w}}(N;X)=-\rho(\theta^{-1}-1)\mathfrak{S}(N)\frac{X}{\log X}\mathcal{J}_{\mathbf{w}}(N/X)+o\left(\frac{X}{\log X}\right)$$

outside an exceptional set of size $o(X)$.

On the support of $w_1(n/X)$, prime powers p^k, $k\ge 2$, contribute $O_{\mathbf{w}}(X^{1/2}(\log X)^2)$ for each N, which is negligible. Moreover $\log p=\log X+O_{\mathbf{w}}(1)$. Dividing by $\log X$ and using partial summation gives (7). □

Remark 7.4. With standard bookkeeping, the exceptional set can be given logarithmic power savings. The density-one form is sufficient for separating what is available from what is pointwise.

8. Explicit classification near the Goldbach threshold

Suppose $1/3 \le \theta < 1/2$. Then, apart from squareful targets annihilated by μ,

$$b_{\theta,X} = -\mathbf{1}_{\mathbb{P}} + \mathbf{1}_{\mathbb{P}_{2,\theta}},$$

where $\mathbb{P}_{2,\theta}$ consists of products rs of two distinct primes exceeding X^{θ}. Consequently the two associated graded columns are

factor degree	additive correlation	coefficient
1	$-\{p+q=N\}$	-1
2	$+\{p+rs=N\}$	$\log((1-\theta)/\theta)$.

Their difference has coefficient

$$-\rho(\theta^{-1}-1) = -1 + \log\frac{1-\theta}{\theta} < 0.$$

At $\theta = 1/3$, Theorem 1.2 therefore gives, in a smooth dyadic form and for almost all even N,

$$S_{\mathbf{w}}(N) - R_{\mathbf{w}}(N) \sim -(1-\log 2)\mathfrak{S}(N)\frac{X}{(\log X)^2}\mathcal{J}_{\mathbf{w}}(N/X). \tag{34}$$

Thus the prime target dominates the rough semiprime target by a fixed proportion for almost all N. This is stronger than mere density-one Goldbach existence, although it is proved by the same mean-square mechanism.

For $1/4 \le \theta < 1/3$, a third, negative column enters. Its coefficient is $-I_3(\theta)$, with I_3 given by (24); the total remains negative because it equals $-\rho(\theta^{-1}-1)$. At each reciprocal threshold $1/k$, one new factor-number column appears. The homological classification is therefore finite on every fixed-θ chamber.

9. Dispersion, L-functions, and the pointwise obstruction

9.1. **What averaging buys.** The density-one proof uses

$$\sum_N |\mathcal{E}_{\mathfrak{m}}(N)|^2 = \int_{\mathfrak{m}} |S_\Lambda(\alpha)S_\theta(\alpha)|^2\,\mathrm{d}\alpha,$$

followed by a supremum bound for the prime sum. Removing the exceptional set would require the pointwise estimate

$$\sup_{N\in[X,2X]}\left|\int_{\mathfrak{m}} S_\Lambda(\alpha)S_\theta(\alpha)\mathrm{e}(-N\alpha)\,\mathrm{d}\alpha\right| = o_\theta\left(\frac{X}{\log X}\right). \tag{35}$$

The elementary $L^2\times L^2$ bound is only $O(X)$, too large by a logarithmic factor. Thus (35) needs cancellation in the interaction of the two sums, not separate norm bounds.

At $\theta = 1/3$, the first factor-number column is the binary Goldbach minor-arc integral itself. The second column is bilinear and is more amenable to dispersion, but no differential in the finite incidence spectral sequence automatically cancels the first column. A new analytic chain homotopy would amount precisely to a twisted prime–Möbius estimate.

9.2. **Character decomposition and reciprocal L-functions.** For $\Re s > 1$,

$$\sum_{P^-(m)>z} \frac{\mu(m)\chi(m)}{m^s} = \frac{1}{L(s,\chi)} \prod_{\ell \le z} \left(1 - \frac{\chi(\ell)}{\ell^s}\right)^{-1}. \tag{36}$$

Analyzing (36) directly requires lower bounds or negative moments for $L(s,\chi)$. GRH places zeros on the critical line but does not by itself provide the uniform negative-moment and character-average control needed for (35).

The finite factor filtration replaces (36), for fixed θ, by the finite family (17). This avoids reciprocal L-functions in smoothed and mean-square problems: large-sieve estimates apply to positive moments of prime Dirichlet polynomials. When one tries to totalize the filtration pointwise, however, the reciprocal L-function difficulty reappears.

Recent work on diagonal Elliott–Halberstam problems twisted by the Möbius function makes this connection explicit; see [2]. Even related fixed-shift Liouville convolutions remain delicate; compare [7].

9.3. **A sufficient pointwise hypothesis.** It is useful to isolate the missing input.

Conjecture 9.1 (Rough prime–Möbius minor-arc estimate)**.** *For every fixed $\theta > 0$ and smooth w_1, w_2, (35) holds.*

Proposition 9.2. *Conjecture 9.1, together with the major-arc estimates of Section 7, implies*

$$\mathcal{G}_{\theta,\mathbf{w}}(N;X) = -\rho(\theta^{-1} - 1)\mathfrak{S}(N)\frac{X}{(\log X)^2}\mathcal{J}_{\mathbf{w}}(N/X) + o\left(\frac{X}{(\log X)^2}\right)$$

uniformly for even $N \in [X, 2X]$.

Proof. Combine Proposition 7.2 with Conjecture 9.1, then remove the von Mangoldt weight as in the proof of Theorem 1.2. □

Corollary 9.3. *At $\theta = 1/3$, Conjecture 9.1 implies the binary Goldbach conjecture for all sufficiently large even integers.*

Proof. The main coefficient is $-(1 - \log 2) < 0$. By (8), negativity implies that the number of prime–prime representations is positive. □

10. What the homological formulation contributes

The construction has four concrete effects.

(1) It gives an exact complex whose Euler characteristic is the signed rough correlation, rather than using "homological" as an analogy.
(2) Its E_1-page separates congruence incidence conditions, while its E_2-page proves that full inclusion–exclusion is exact.
(3) The factor-number filtration replaces a reciprocal L-function by finitely many prime-product strata for fixed θ. This is useful for large-sieve and mean-square estimates.
(4) The filtration localizes the pointwise obstruction. In the chamber $1/3 \le \theta < 1/2$, it is the first, prime–prime column; higher columns do not conceal an algebraic differential that would solve Goldbach.

What it does not provide is a finite-field-style weight theorem over the archimedean height variable. Over a finite field, Frobenius traces and Deligne-type bounds can turn nontrivial cohomology into square-root cancellation. For fixed integers N, the corresponding trace estimate is exactly the missing pointwise harmonic-analysis problem.

11. Further directions

Several extensions are realistic.

(1) Write a quantitative exceptional-set version of Theorem 1.2, with logarithmic or power savings.
(2) Study the same signed correlation in arithmetic progressions in N. The bar E_1-page is naturally compatible with a Barban–Davenport– Halberstam variance.
(3) Use dispersion and spectral large-sieve estimates on the $j \geq 2$ factor columns, and measure precisely how much of Conjecture 9.1 remains in the $j = 1$ column.
(4) Extend the construction to rings of integers of number fields. The parity line then belongs naturally to squarefree ideals. Units, class groups, nonprincipal prime ideals, archimedean regions, and local admissibility enter the analytic statement, but the incidence complex itself is unchanged.
(5) Compare the Dickman identity with Poisson–Dirichlet limits and with the exact finite-n symmetric-group identity. This may lead to secondary terms that match the $1/\log X$ expansion on the integer side.

12. Logical status of the statements

Statement	Status	Main input
Homological resolution and E_2-degeneration	proved	finite Koszul/bar complexes
Signed Dickman identity	proved	boundary recurrence and Dickman equation
Smoothed global mean	proved	prime number theorem and factor strata
Density-one signed Goldbach asymptotic	unconditional	Siegel–Walfisz, major arcs, Vaughan estimate, Parseval
Uniform pointwise asymptotic	conditional	Conjecture 9.1
Binary Goldbach consequence at $\theta = 1/3$	open	pointwise minor-arc cancellation

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